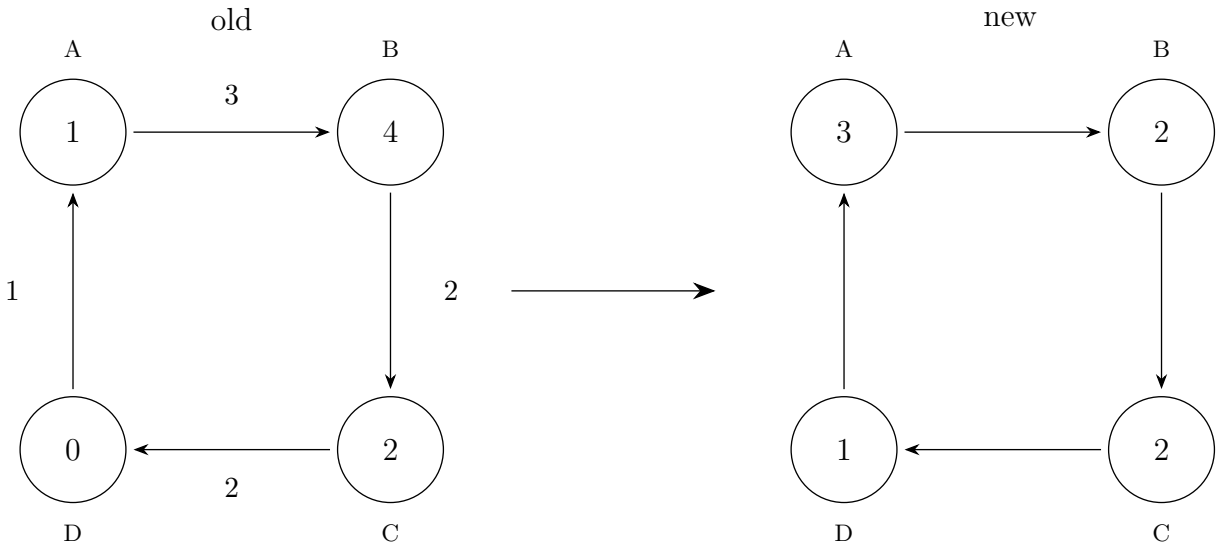
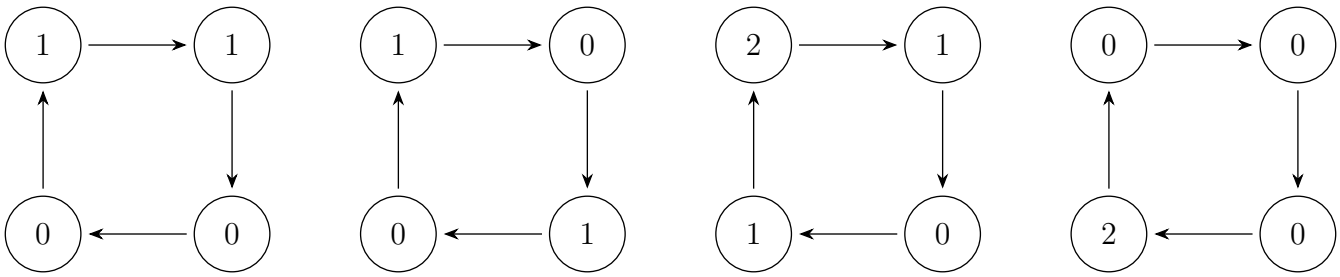
Use whole-number heights, including 0. Follow each arrow: the gap between its two old heights becomes the new height at its tail. Make every new tower before changing any old tower. An all-zero start takes 0 moves.

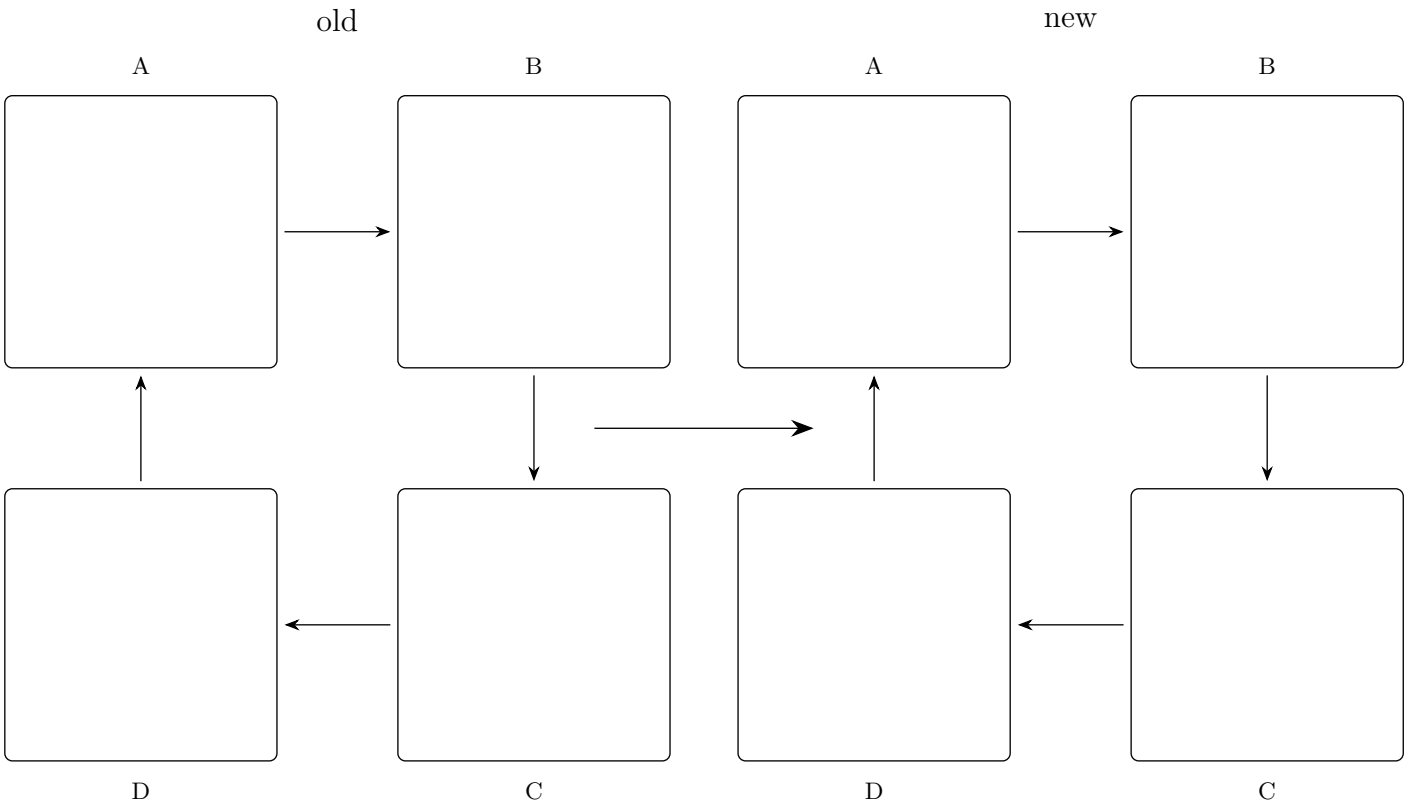


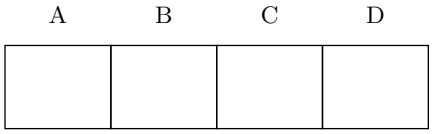

Edge A to B has gap 3, so new A is 3.

**Problem 1:** Build these starts and keep making new rings. Stop when all four towers are 0.

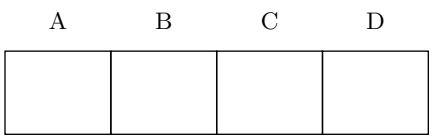


**Problem 2:** Choose heights from 0 to 4. Find a start that lasts as many moves as you can; count the move that makes all four towers 0.



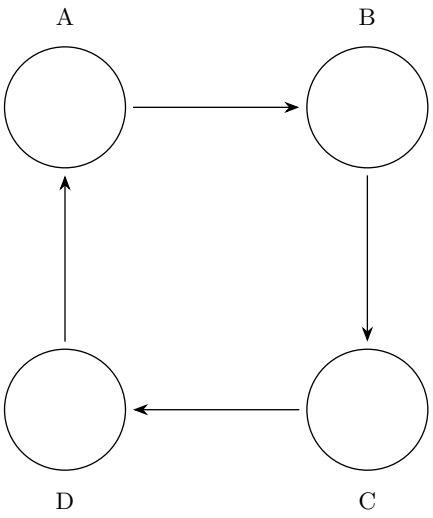



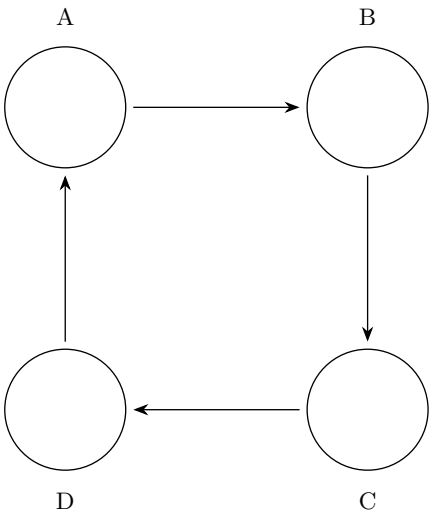

Moves to zero: _____



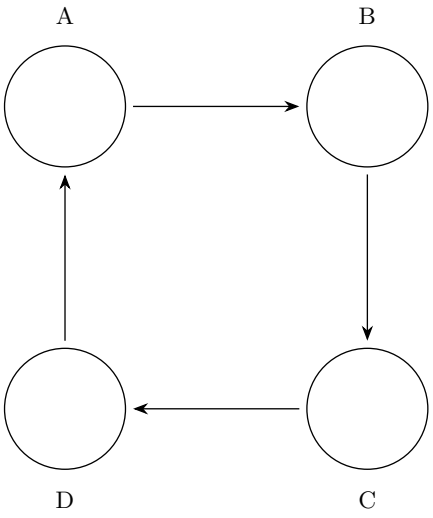

Moves to zero: _____

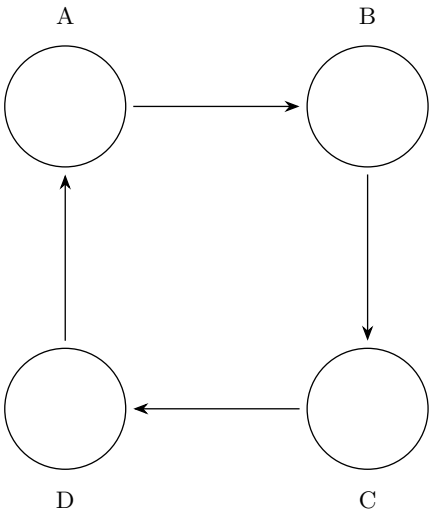
**Problem 3:** Find different starts that become all zeros in one move.



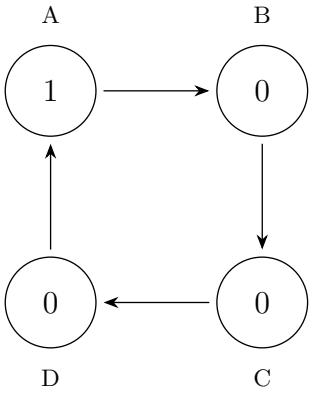




**Problem 4:** Find different starts that first become all zeros on the second move.

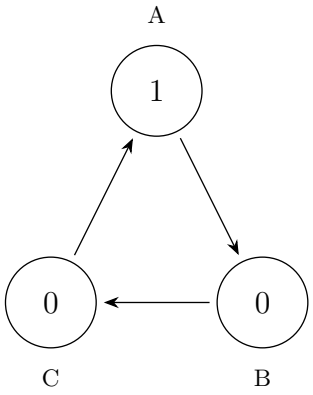


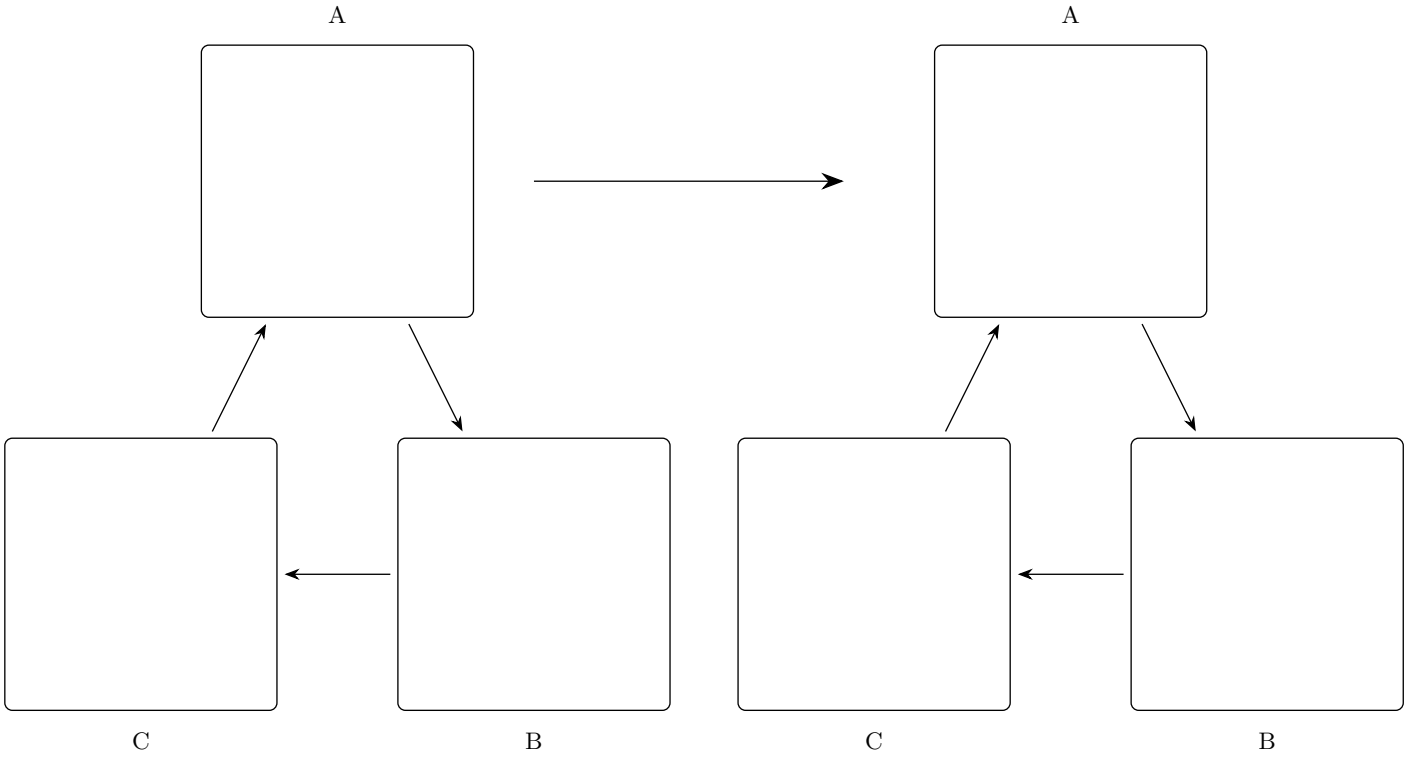
**Problem 5:** Try both starts with the same gap rule. Stop at all zeros or when the whole ring repeats.




old



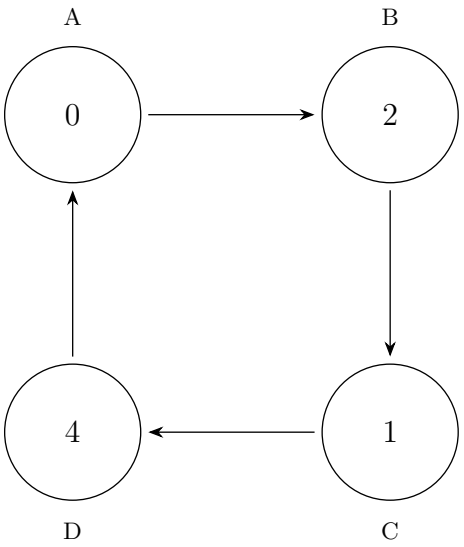

new

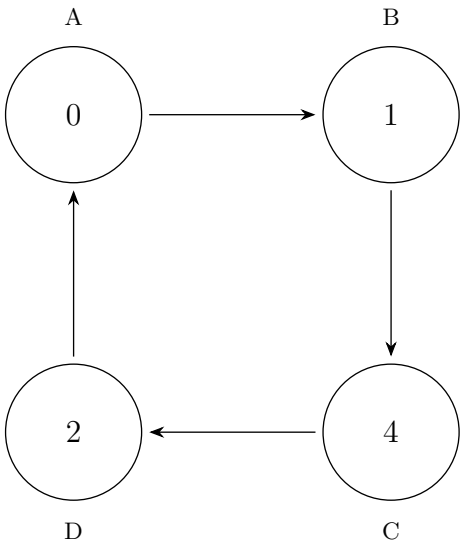


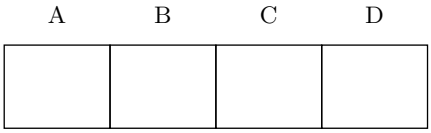

| | A | B | C |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

| | A | B | C |
|---|---|---|---|
| 4 | | | |
| 5 | | | |
| 6 | | | |

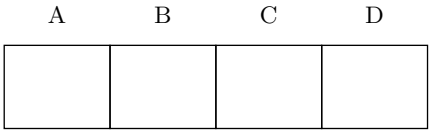
**Problem 6:** Put 0, 1, 2, and 4 around the four stations. Find an order that lasts longer than the other orders you try.



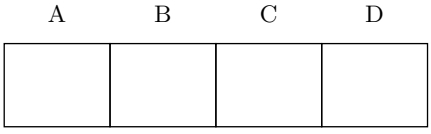




Moves to zero: _____




Moves to zero: _____




Moves to zero: _____